

MTH 377/577 Convex Optimization

Problem set 5

1. Which of the following sets are (1) Affine; (2) Convex; (3) Closed; (4) Bounded
 - (a) $A = \mathbb{R}^3 \setminus \{(0, 0, 0)\}$. Ans. None. Show with counterexamples.
 - (b) $B = \{(x_1, \dots, x_n) \in \mathbb{R}^n | x_i > 1, \forall i \leq n\}$. Ans. None.
 - (c) Let $A_k = \{(x_1, \dots, x_k) \in \mathbb{R}^n | x_i = \frac{1}{k}, \forall i \leq k\}$. Suppose for some $n \geq 1$, $A = \cup_k A_k$, $k \leq n$. Which of the 4 properties mentioned above does A satisfy? Ans. Closed, Bounded.
 - (d) $D = A \setminus B$ where $A = \{(x_1, \dots, x_n) \in \mathbb{R}^n | x_i \leq 5\}$ and $B = \{(x_1, \dots, x_n) \in \mathbb{R}^n | 4 < x_i \leq 5\}$. Ans. Convex, closed and bounded.
2. A polyhedron is always convex. True/False. Ans. True. It is the intersection of a finite number of hyperplanes and halfspaces (write down expression for the same- ref slides).
3. Show that a convex cone is a subspace. Ans. Let K be a convex cone (write definition). Closed over addition, scalar multiplication, contains zero vector (show).
4. Suppose $x = (x_1, \dots, x_n)$, $x_i < 0$ for all $i \leq n$ is a solution to the system $Ax = b$. Can the Farkas' Lemma help you for the feasibility of this solution? Ans. No. Farkas' Lemma is applicable for non-negative solutions. The sign of Ax changes when $x < 0$, and the condition $yA \geq 0$, $yb < 0$ becomes stricter than required for the existence of the solution.
5. Convert the following LP into standard form, write down its dual and

check if a non-negative solution exists:

$$\begin{array}{ll} \max & 3x_1 + 2x_2 \\ \text{such that} & 2x_1 + x_2 \leq 6 \\ & x_1 + x_2 \leq 3 \\ & 2x_1 \leq 4 \\ & x_1, x_2 \geq 0 \end{array}$$

Ans. Standard form:

$$\begin{array}{ll} \max & 3x_1 + 2x_2 \\ \text{such that} & 2x_1 + x_2 + s_1 = 6 \\ & x_1 + x_2 + s_2 = 3 \\ & 2x_1 + s_3 = 4 \\ & x_1, x_2, s_1, s_2, s_3 \geq 0 \end{array}$$

Write down the dual and use Farkas Lemma (do on your own) with

$$A = \begin{bmatrix} 2 & 1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 \\ 2 & 0 & 0 & 0 & 1 \end{bmatrix}$$

6. Suppose for an optimization problem f_0 , strong duality does not hold. Can you still use Lagrangian function to solve the problem? Ans. no. the optimal values x^* cannot be recovered from Lagrangian dual function since the following condition may no longer hold:

$$f_0(x^*) = g(\lambda^*, v^*) = \inf_{x \in D} L(x, \lambda^*, v^*) \leq L(x, \lambda^*, v^*)$$

7. Give an example of an optimization problem with non-linear constraints and demonstrate how you can convert them to linear and solve using KKT. (Ans. See Principal Agent model, Lecture 12).
8. An envy free allocation is a fixed point. True/False? Ans. True. Show using Sperner's Lemma and the triangulation procedure for 3 agent case as discussed in class. (Key argument: for any equilateral triangle, a triangulation exists such that each edge can be assigned a distinct label. Construct a sequence of such triangles (A_k) . As $k \rightarrow \infty$, A_k converges to a vector in R^3).
9. Consider the following optimization problem:

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{s.t.} & f_i(x) \leq u_i, i = 1, 2, \dots, m \\ & h_i(x) = v_i, i = 1, 2, \dots, p \end{array}$$

where $x \in R^n$ Suppose strong duality holds and f_0, f_i, h_i are all continuously differentiable. Suppose u_i decreases by $\alpha > 0$. How will the optimal value of f_0 change? Ans. The optimal value p^* will increase since $\lambda_i^* = \frac{-\delta p^*(0,0)}{\delta u_i}$; $v_i^* = \frac{-\delta p^*(0,0)}{\delta v_i}$ See Lecture 15 (write all steps).